Problem 9.15

Treasury bills, or T-bills, are sold in terms ranging from a few days to 26 weeks. Bills are sold at a discount from their face value. For instance, you might pay $990 for a $1,000 bill. When the bill matures, you would be paid $1,000. The difference between the purchase price and face value is interest. A bill for 100 is purchased for 96 three months before it is due. Find:

(a) The nominal rate of discount convertible quarterly earned by the purchaser.

(b) The annual effective rate of interest earned by the purchaser.

Solution

The face value of the T-bill is 100 and the purchase price is 96. Since the bill matures in three months, this corresponds to one quarter of a year.

Part (a)

Let $d^{(4)}$ be the nominal rate of discount convertible quarterly. Since the purchase price is 96 and the maturity value is 100, we have

$$96 = 100\left(1 - \frac{d^{(4)}}{4}\right)^1.$$

Dividing both sides by 100 gives

$$0.96 = 1 - \frac{d^{(4)}}{4}.$$

Therefore,

$$\frac{d^{(4)}}{4} = 0.04,$$

so

$$d^{(4)} = 0.04(4) = 0.16.$$

Thus, the nominal rate of discount convertible quarterly is

$$\boxed{16\%}.$$

Part (b)

The annual effective interest rate i is equivalent to the quarterly discount rate found above. Hence,

$$1 + i = \left(1 - \frac{d^{(4)}}{4}\right)^{-4}.$$

Since $1 - d^{(4)}/4 = 0.96$, we get

$$i = (0.96)^{-4} - 1.$$

Thus,

$$i \approx 0.177376.$$

Therefore, the annual effective rate of interest earned by the purchaser is

$$\boxed{17.7376\%}.$$